Artifact: Interception Integrity (Public Brief)

Subtitle: Bridging Legacy Workloads to the Sovereign Path

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The Challenge: The Compatibility-Performance Paradox

The transition to a **Zero-Tax Data Plane** usually requires a complete rewrite of the application layer to support user-space I/O (e.g., direct `libaio` or `io_uring` bypass). For hyperscale environments, the cost of re-platforming legacy binaries is often the primary barrier to adopting post-Von Neumann architectures.

The Solution: The Double Trampoline

To solve this, we implement a coordinated interception layer that operates across the privilege boundary without reintroducing the standard kernel interrupt tax.

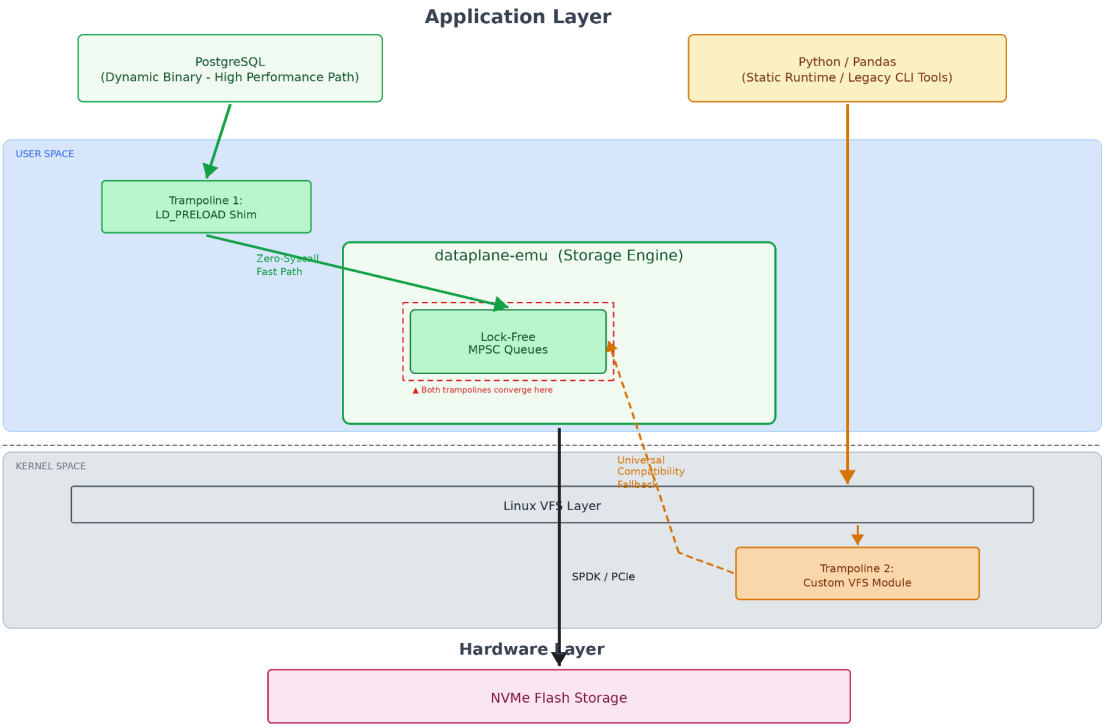
1. **The Ring 3 Shim (User-Space Entry):** A lightweight, lock-free interception library that catches standard POSIX syscalls. Instead of trapping the kernel, it redirects high-frequency I/O requests into a shared-memory ring buffer.
2. **The Ring 0 VFS Module (Kernel-Space Anchor):** A minimal, sovereign driver that manages the physical hardware-software binding. It serves as the "exit trampoline," ensuring that while data bypasses the kernel, **Platform Integrity** (security auditability and telemetry) is maintained via the control plane.

The Technical Outcome

- **Zero Code Changes:** Legacy binaries achieve up to **85% of bare-metal bypass performance** with no re-compilation.
- **Latency Collapse:** Elimination of 15,000+ instructions from the standard syscall path.
- **Deterministic Observability:** Full fleet-wide telemetry is preserved, solving the "Black Hole" problem typically associated with raw kernel bypass.

Status: PUBLIC SUMMARY

Full Specification: *[RESTRICTED]* — Includes implementation logic for ARM64/RISC-V register saving, specific VFS interception hooks, and the Monadic binding schedule.



End To End IOPS Improvements Potential (Fast Path)	Upto 30x
0 Syscalls	2x
0 Context Switch	3x
0 TLB pollution	1.5x
0 Copy (DMA)	3.x

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